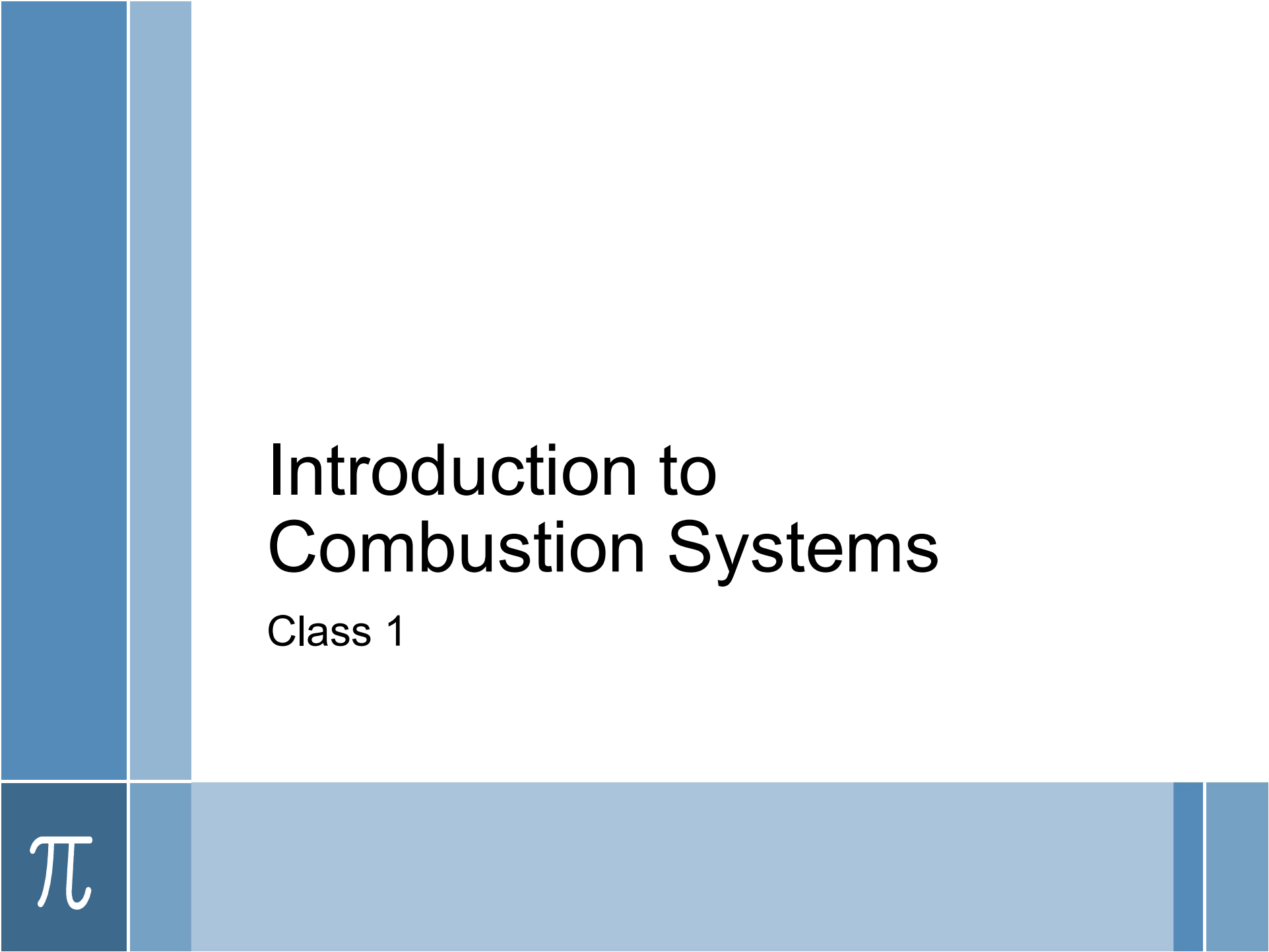


Introduction to Combustion Systems

Class 1

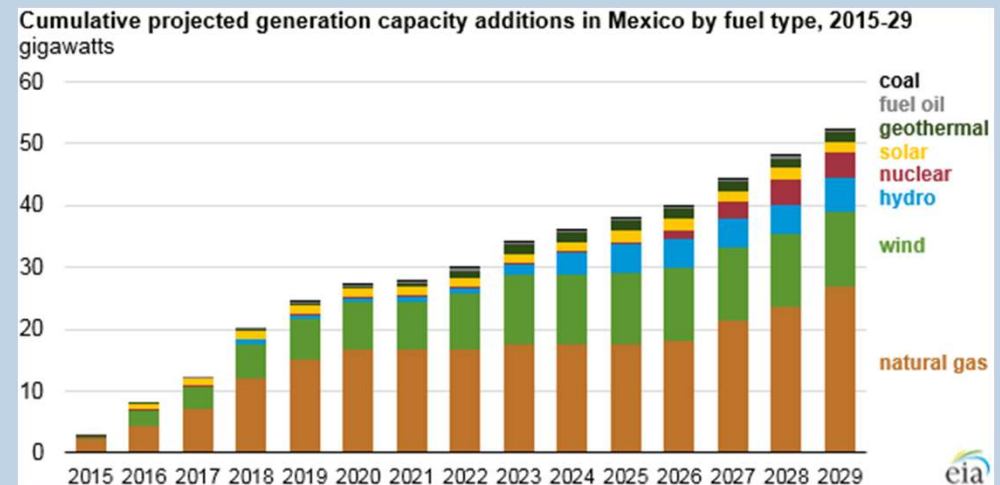
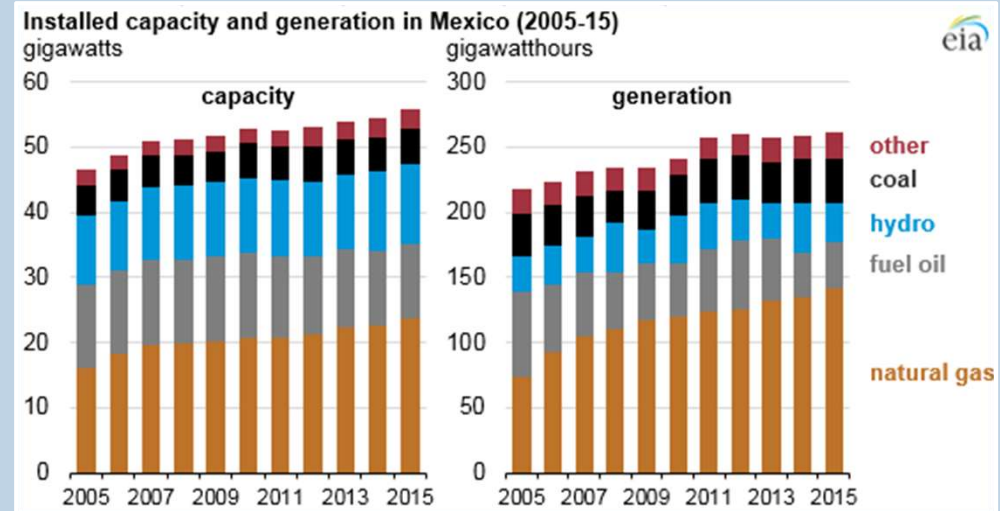


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WHY STUDY COMBUSTION

Natural gas is Mexico's largest source of electricity generation, accounting for 54% of the country's generation in 2015, up from 34% in 2005. According to Mexico's national energy ministry (SENER), more than 60% of Mexico's electric capacity additions between 2016 and 2020 are projected to come from natural gas-fired power plants, and significant natural gas capacity additions are expected to continue through 2029.

Aircraft are entirely powered by on-board fuel burning, and most cars are gasoline or diesel engine driven. Among other industrial processes that rely heavily on combustion.





A definition of Combustion

- › Webster's dictionary provides a useful starting point for a definition of combustion as *“rapid oxidation generating heat, or both light and heat; also, slow oxidation accompanied by relatively little heat and no light.”*



Thermodynamics and Chemistry Review

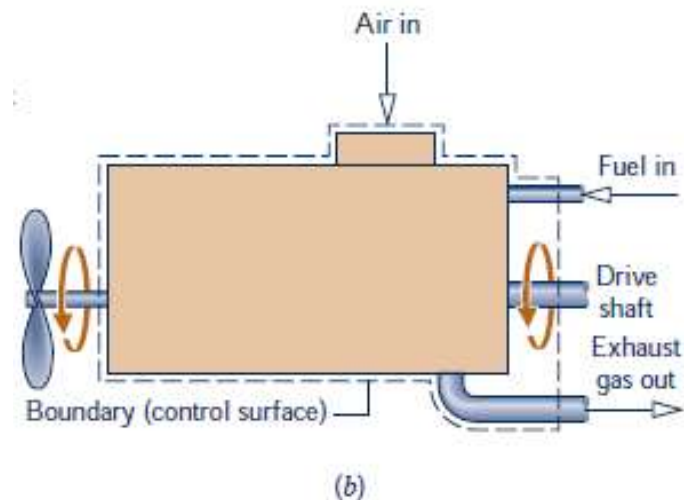
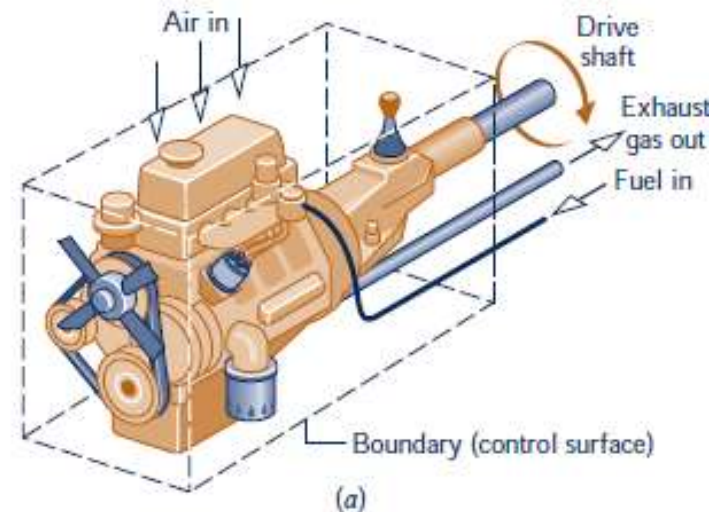


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Definition of system...

- › Helps us identify the body we are studying.
 - Boundary
 - › Imaginary or tangible
 - › At rest (static) or in motion
 - Surroundings

- › Types of systems:
 - Closed system
 - › Have a fixed quantity of matter
 - Control volume
 - › Matter can be exchanged
 - Isolated system



Property, state and process

› Property

- Characteristics which describes the system with a numerical value at a given time.

› State

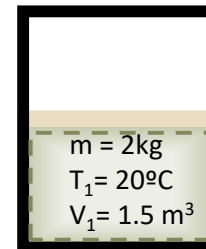
- Condition of the system
- Described by its properties.

› Process

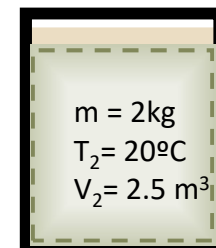
- Transformation from one state to another

› Steady state

State 1



State 2





Energy

- › Capability of doing work
- › Total energy of a system (E_T):
 - Sum of all the different energy forms
 - Depending on the coordinate frame:
 - › Macroscopic energy
 - › Microscopic energy
- › Internal energy (U)
 - Chemical: molecule bonds
 - Latent: change of state
 - Sensible: change of temperature
- › Energy transfer
 - Work, heat and mass

$$E_T = E_k + E_p + U$$

Work, Power and Heat Transfer

› Work (W)

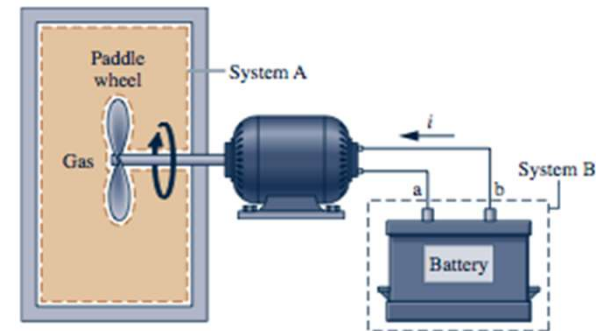
- A means to transfer energy
- Force acting through a distance
 - › $W > 0$ done by the system

› Power

- Rate at which energy transfer by work occurs (W)

› Heat transfer(Q)

- Means to transfer energy
- Because of a temperature difference
 - › $Q > 0$ heat transfer to the system



Examples of work

1st law of thermodynamics

“Energy cannot be created or destroyed but it can be transformed”

› Defines the system's property:

- Total Energy change (E_T).
 - › The difference between two states

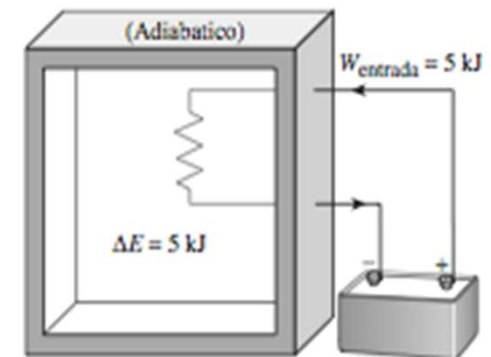
› Introduces the idea of energy balance:

- There cannot be losses!

General energy balance

Net change in the amount of energy within the system during a process	=	Net amount of energy transferred in the system during a process OR final state	—	Net amount of energy transferred out the system during a process OR initial state
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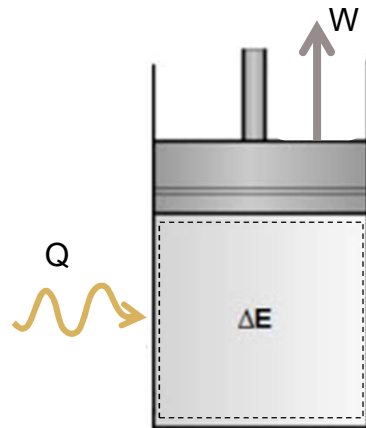
$$\Delta E_T = Q - W + m$$



$$\begin{aligned}\Delta E_T &= -W_{ad} \\ E_{state_2} - E_{state_1} &= -W_{ad_2} + W_{ad_1}\end{aligned}$$

Energy balance

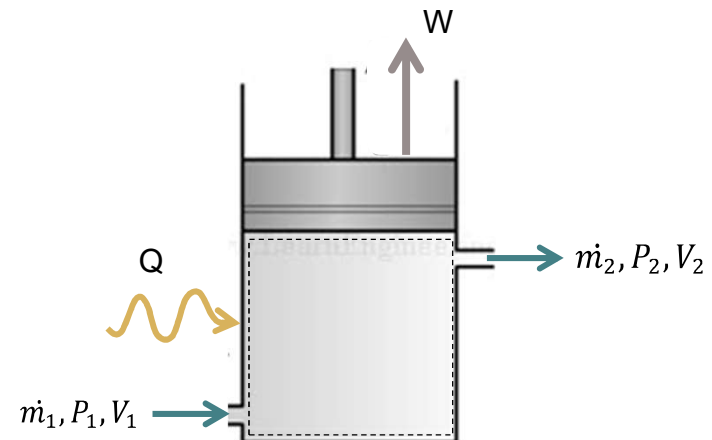
Closed system



Change in total energy would be:

$$\Delta E_{\text{tot}} = Q - W$$

Control volume
(open system)



$$\frac{dE_{\text{tot}}}{dt} = \dot{Q} - \dot{W} + \dot{m}_1(E_U + E_P + E_K) - \dot{m}_2(E_U + E_P + E_K)$$

2ND LAW OF THERMODYNAMICS

Spontaneous processes:

Happen in a definite direction

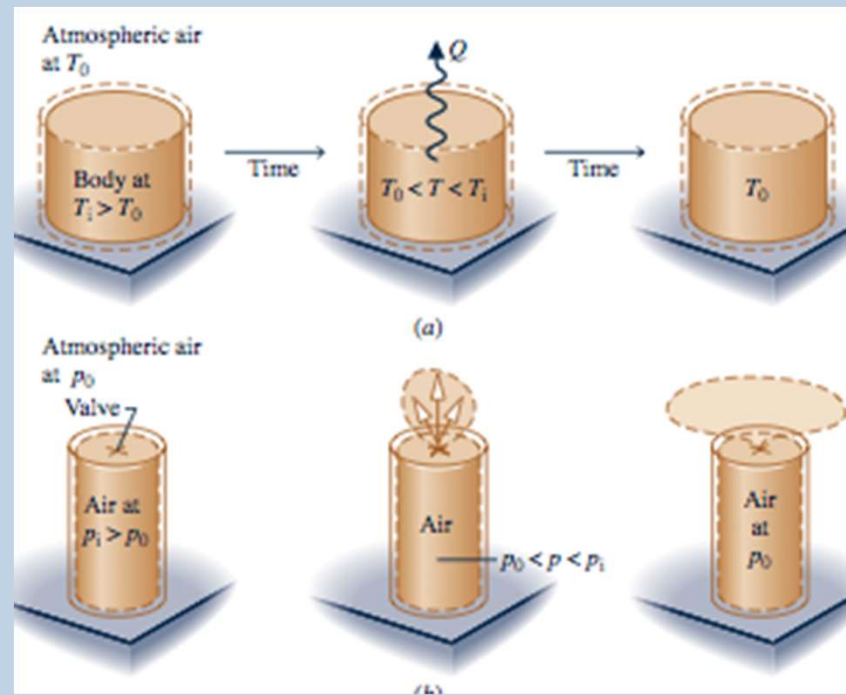
Until equilibrium is achieved

Internally or with surroundings

Not every process that follows the 1st law can occur.

Energy has quantity, direction and quality.

Higher the energy temperature higher the energy quality.



2nd law of thermodynamics: Entropy

› Entropy (S):

- Energy lost in the form of heat transfer at the boundary system.
 - › Property based on Clausius inequality
 - › Disorder or uncertainty of the system
- Generated by the presence of irreversibilities

$$\oint \left(\frac{\delta Q}{T} \right)_b \leq 0$$

Clausius' inequality

› Property of the system:

- › Measure a change in entropy of internally reversible processes

› Irreversibilities:

- Factors that enable energy loss
- Friction, expansion of gas/liquids, heat transfer at finite temperature...

$$\Delta S = S_2 - S_1 = \int_1^2 \left(\frac{\delta Q}{T} \right)_{int \ rev}$$

Definition of entropy change

Entropy balance

› Entropy balance for a general system is expressed:

- Entropy can be generated

$$\underbrace{S_{in} - S_{out}}_{\substack{\text{Net amount of} \\ \text{entropy transferred} \\ \text{across the boundary}}} + \underbrace{S_{gen}}_{\substack{\text{Generated/produced} \\ \text{amount of entropy} \\ \text{within the system}}} = \underbrace{\Delta S_{sist}}_{\substack{\text{Net entropy change in} \\ \text{the contained amount} \\ \text{within the system}}} = S_2 - S_1 = \int_1^2 \left(\frac{\delta Q}{T} \right)_b + m_{in}S_{in} - m_{out}S_{out} + S_{gen}$$

- Entropy can be produced due to irreversibilities
- Entropy can be transferred through: mass and heat transfer

› Entropy accompanying heat transfer (S_Q)

- Sign convention applies

$$S_Q = \int_1^2 \left(\frac{\delta Q}{T} \right)_b$$

› Entropy accompanying mass flow

$$S_m = m_{in}S_{in} - m_{out}S_{out}$$

Exergy

› Exergy (E)

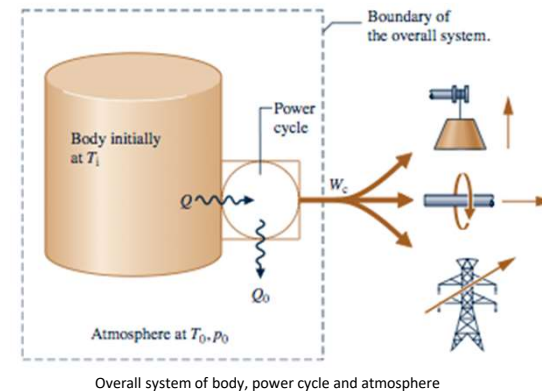
- Property quantify the potential for use.
- The potential for developing work
- Destroyed by irreversibilities
- Based on the second law.

› Environment

- Large enough to remain uniform
 - › Default: $T_0 = 25\text{ °C}$, $p_0 = 1\text{ atm}$
 - › Has no irreversibilities

› Dead state

- System reaches equilibrium with environment
 - › Overall system is in equilibrium
 - › No potential for developing work



“ Exergy: the maximum theoretical work obtainable from an overall system ”

Exergy Balance

› Exergy of a given state

$$E = (U - U_0) + p_0(V - V_0) - T_0(S - S_0) + E_p + E_k$$

U, V, S – variables of the system at given state

U_0, V_0, S_0, p_0, V_0 – variables of the system at dead state

E_p, E_k – variable relative to environment

› Exergy change

$$E_2 - E_1 = (U_2 - U_1) + p_0(V_2 - V_1) - T_0(S_2 - S_1) + (E_{p_2} - E_{p_1}) + (E_{k_2} - E_{k_1})$$

› Exergy can be transferred by:

› Heat transfer (E_q)

$$E_q = \int_1^2 \left(1 - \frac{T_0}{T_b}\right) \delta Q$$

› Work (E_w)

$$E_w = W - p_0(V_2 - V_1)$$

› Mass flow (E_m)

$$E_m = m_{in} e_{f \cdot in} - m_{out} e_{f \cdot out}$$

Specific flow exergy

$$e_{f \cdot i} = h - h_0 - T_0(S - S_0) + E_p + E_k$$

$$E_D = T_0 S_{gen}$$

› Exergy destruction (E_d)

› Exergy balance:

$$\Delta E_{syst} = E_2 - E_1 = E_q - E_w + E_m - E_D$$



Mole and molecular weight

> Mole

- Measurement unit for Atoms, molecule or particles
 - Analogue to pair or dozen
 - Based on Avogadro's number

> Mole number (N_i)

- How many moles you have

> Molecular weight (MW_i)

- Weight in grams of a mole of something
 - > Elements: atomic weight
 - > Compounds:
 - Known the compound's formula
 - Know the MW of the elements

1 mole = 6.02×10^{23} atoms
Molecules
particles

1 mole He $\rightarrow$ 6.02×10^{23} atoms of He
1 mole H_2 $\rightarrow$ 6.02×10^{23} molecules of H_2
1 mole H_2O $\rightarrow$ 6.02×10^{23} molecules of H_2O

$N_{H_2} = 3 \rightarrow$ 3 moles of
 H_2

$$MW_i = \frac{\text{grams}}{\text{mole}} = \frac{\text{grams}}{\text{g} \cdot \text{mole}} = \frac{\text{lb}}{\text{lb} \cdot \text{mole}}$$



Examples

- › What is the molecular weight for amonia NH_3 ?
 - g/mol
 - lb/lb-mol
 - Kg/k-mol
- › There are 38 kg of amonia, what is the equivalent in moles?
- › If there is 4 kmol of amonia, how many pounds of amonia does it represent?

Composition of a Mixture

- › Mass of a mixture (kg – lbm)

$$m_{mezcla} = \sum_{i=1}^k m_i$$

- › Molar number of a mixture (moles)

$$N_{mezcla} = \sum_{i=1}^k N_i$$

- › Mass Fraction

$$Y_i = \frac{N_i}{N_{mezcla}} = \frac{\text{moles } i}{\text{moles de mezcla}}$$

- › Mass Fraction

$$X_i = fm_i = \frac{m_i}{m_{mezcla}} = \frac{\text{kg } i}{\text{kg mezcla}}$$

- Mass of a substance

- $m = N \cdot PM$

$$\sum_{i=1}^k Y_i = 1$$

$$\sum_{i=1}^k X_i = 1$$

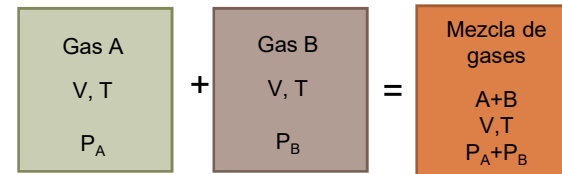
› Molecular Weight of a Mixture:

- Relationship between y_i y x_i :

$$PM_{mezcla} = \sum_i y_i \cdot PM_i$$

$$y_i \xrightleftharpoons[PM_i/PM_{mezcla}]{PM_{mezcla}/PM_i} x_i$$

- Partial pressure of a mixture:
 - Relationship with the volumetric fraction
- For ideal gases



$$P_m = \sum_{i=1}^k P_i (T_m, V_m)$$

$$\frac{P_i}{P_m} = \frac{N_i R_u T_m / V_m}{N_m R_u T_m / V_m} = \frac{N_i}{N_m} = y_i$$

$$\bar{h}_{mezcla} = \sum_i y_i \cdot \bar{h}_i$$